

E-WATER DISTILLATION SYSTEM: AN ARDUINO MEGA 2560-BASED PORTABLE WATER QUALITY MONITORING AND DISTILLATION APPARATUS FOR SAFE DRINKING WATER PRODUCTION

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Abstract— Access to safe and potable water remains a critical public health challenge in the Philippines, particularly in rural communities where waterborne disease transmission is exacerbated by inadequate water treatment infrastructure. This paper presents the design, development, and evaluation of a Portable E-Water Distillation System that employs a nichrome wire heating element integrated with an Arduino Mega 2560 microcontroller and a suite of water quality sensors — including pH, turbidity, Total Dissolved Solids (TDS), temperature, and water level sensors — to automate the distillation process and continuously monitor the quality of distilled water output. The system was evaluated using tap water collected from the College of Engineering and Architecture at the Polytechnic University of the Philippines (PUP-CEA). Experimental results demonstrated that following a single-pass distillation cycle, the system achieved a turbidity reading of 4.7 NTU, a pH of 6.8, and a temperature of 26°C, all within the acceptable limits prescribed by the World Health Organization (WHO). However, the TDS measurement of 741 ppm remained above the WHO minimum acceptable concentration of 500 mg/L, indicating that a single thermal distillation pass is insufficient to reduce dissolved solids to potable standards. A closed-loop feedback mechanism was successfully implemented, automatically recycling water that did not meet quality thresholds back to the distillation chamber. The system demonstrated functional sensor integration, real-time parameter monitoring via a 16×2 LCD display, and automatic dispensing control through relay-actuated water pumps. Key limitations identified include extended distillation cycle time, heating element overheating risk, and TDS reduction insufficiency. Future improvements include multi-stage filtration, solar power integration, and chemical disinfection supplementation..

Keywords— water distillation system, Arduino Mega 2560, water quality monitoring, TDS, turbidity, pH sensor, IoT, embedded systems, safe drinking water,

I. INTRODUCTION

Water is an indispensable resource for human life, economic productivity, and ecological balance. Its quality directly impacts the health and well-being of individuals and communities worldwide [1]. The World Health Organization

(WHO) estimates that approximately 2.2 billion people globally lack access to safely managed drinking water services, and contaminated water sources are causally linked to the transmission of severe waterborne diseases including cholera, typhoid fever, hepatitis A, dysentery, and poliomyelitis. The WHO attributes approximately 500,000 deaths annually to diarrheal diseases alone, the majority of which are preventable through adequate water treatment and sanitation [8].

In the Philippines, the challenge of securing clean, potable water is particularly acute in rural and peri-urban communities. A socioeconomic study conducted in an urban poor community in the Philippines revealed that 65.3% of households (257 out of 393 surveyed families) relied on filtered or distilled water from commercial water refilling stations, while 33.1% of families consumed tap water whose safety was contingent on the reliability and management practices of local water utility providers [5]. This dependency on external water supply chains exposes households to significant health risks, particularly during supply disruptions or infrastructure failures.

Manual water quality monitoring presents operational challenges in terms of cost, time, and scalability. Conventional water quality assessment requires laboratory-grade instruments and trained personnel, making real-time field monitoring impractical for low-resource settings [3]. The emergence of low-cost embedded sensor systems and microcontroller-based platforms has created new opportunities for automated, real-time water quality surveillance at the point of use [1][8].

Contemporary water treatment infrastructure, particularly in developing nations, requires ongoing improvements to address the growing spectrum of environmental contaminants documented in source water and distribution networks [6]. Research conducted in the United States has shown that even in developed urban systems, tap water monitoring gaps exist, resulting in potential undetected exposure to chemical contaminants [7]. In the Philippine context, the use of household water storage tanks further

introduces risks of microbiological and chemical contamination during storage and distribution [9].

To address these systemic challenges, the research team developed a Portable E-Water Distillation System that integrates thermal water distillation with automated, sensor-based quality assessment. The system is designed to process tap water through filtration and distillation stages, continuously evaluate the distilled product against WHO standards, and automatically recycle non-conforming water for further treatment. The system is built around an Arduino Mega 2560 microcontroller, which orchestrates data acquisition from multiple sensors and controls actuators including water pumps and the heating element via relay circuits.

The primary objectives of this study are:

1. To design and implement a portable, low-cost water distillation system controlled by an Arduino Mega 2560 microcontroller.
2. To integrate multi-parameter water quality sensors (pH, turbidity, TDS, temperature, and water level) for automated assessment of distilled water quality.
3. To validate the system's distillation and quality evaluation performance against WHO drinking water standards.
4. To implement a closed-loop water recycling mechanism that re-processes non-compliant output water.

II. METHODOLOGY

The methodology encompasses a systematic engineering approach to the development of a portable water distillation and quality monitoring system. The development process was structured into five principal phases: project design, software development, hardware development, system integration, and experimental testing.

1. System Architecture and Conceptual Framework

The E-Water Distillation System operates as a closed-loop embedded control system. Tap water from an external source enters the system and passes through a preliminary mechanical filtration stage to remove coarse particulates before being collected in the primary distillation container. The Arduino Mega 2560 monitors the water level through a resistive water level sensor. Upon detection of sufficient water volume, the relay module activates the nichrome wire heating element to initiate thermal distillation.

Steam produced during the boiling process travels through a condenser tube, where it cools and condenses into the secondary collection container. The water level sensor in the secondary container triggers the activation of the quality monitoring sensors — pH, turbidity, and TDS — once a measurable volume of condensate has accumulated. The microcontroller evaluates sensor readings against programmed WHO threshold parameters. Water meeting all acceptance criteria is automatically dispensed via a 5V water pump through the output port. Water failing any parameter is

re-circulated to the primary distillation container via a recycle pump for reprocessing.

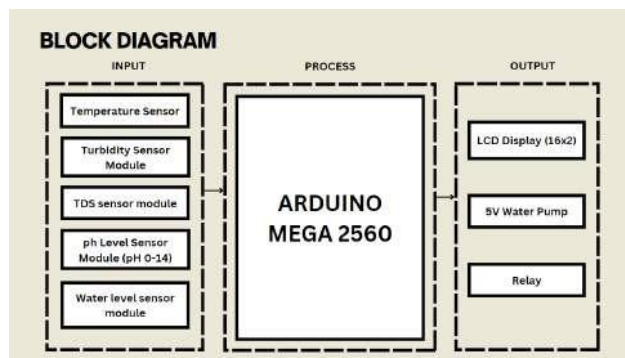


Figure 1: Block Diagram

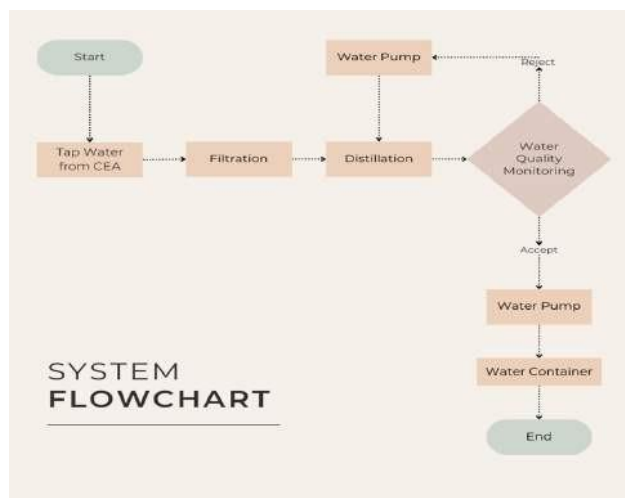


Figure 2: System Flowchart

2. Hardware Design and Component Specifications

The hardware architecture of the E-Water Distillation System integrates sensing, actuation, display, and power management subsystems.

The Arduino Mega 2560 serves as the central processing unit, providing 54 digital I/O pins and 16 analog input channels sufficient to accommodate all sensor and actuator connections without multiplexing. Operating at 5V with a 16 MHz clock frequency, the microcontroller provides adequate computational throughput for real-time multi-sensor data acquisition and decision logic execution.

The nichrome wire heating element was selected for its high electrical resistance, thermal stability, and resistance to oxidation at elevated temperatures. The element is powered through a TRIAC-based dimmer circuit that modulates power delivery to control heating intensity, mitigating the risk of thermal runaway during operation. The relay module — rated for 10A switching at the operating voltage — provides electrical isolation between the low-voltage microcontroller signals and the higher-power heating element and pump circuits.

All sensors were individually calibrated prior to system integration. The pH sensor (range: 0–14, ± 0.1 pH accuracy) was calibrated using standard buffer solutions at pH 4.0, 7.0, and 10.0. The turbidity sensor, which operates on the principle of optical transmittance, was calibrated against known turbidity standards (0–3000 NTU). The TDS sensor employs the relationship between solution conductivity and dissolved solid concentration, calibrated using standard KCl solutions at known concentrations. The DS18B20 temperature sensor ($\pm 0.5^\circ\text{C}$ accuracy over the range -55°C to $+125^\circ\text{C}$) required no field calibration due to its factory-trimmed internal reference.

The complete list of hardware components and their specifications is presented in Table I.

TABLE I. Hardware Components and Specifications of the E-Water Distillation System

Component	Specification	Function
Arduino Mega 2560	ATmega2560, 54 I/O pins, 16 MHz	Central microcontroller; processes sensor data and controls actuators
pH Level Sensor	pH 0–14, ± 0.1 accuracy	Measures acidity/alkalinity of distilled water
Turbidity Sensor	0–3000 NTU range	Detects suspended particles via optical transmittance
TDS Sensor	0–9990 ppm, $\pm 2\%$ accuracy	Measures dissolved ionized solids via conductivity
Temperature Sensor (DS18B20)	-55°C to $+125^\circ\text{C}$, $\pm 0.5^\circ\text{C}$	Monitors water temperature during distillation and output
Water Level Sensor	Resistive/float type	Triggers quality monitoring when distilled container fills
LCD Display (16×2)	16 characters $\times$ 2 rows, I ² C	Real-time parameter display for operators
5V Water Pump	5V DC, centrifugal	Transfers or recycles water based on quality evaluation
Nichrome Wire Heater	High-resistance alloy, 220V AC	Primary heating element for thermal distillation
Relay Module	5V trigger, 10A rated	Controls heater and pump switching via microcontroller
Dimmer Circuit	TRIAC-based	Regulates heater power to prevent overheating

3. Software Design and Control Logic

The control firmware was developed using the Arduino Integrated Development Environment (IDE), which provides a C/C++-based programming environment compatible with the ATmega2560 architecture. Open-source sensor libraries were adapted and integrated into a unified control program for the system.

The relay module was programmed to toggle the heating element at 30-second intervals during the distillation phase, providing a duty-cycle-based thermal management strategy that balances heating efficiency with overheating prevention. This interval was empirically determined during preliminary trials. The control logic implements a state machine with the following principal states: (1) IDLE — system awaiting water input; (2) FILTRATION — mechanical pre-filtration active; (3) DISTILLATION — heater active, temperature monitoring; (4) QUALITY_CHECK — multi-sensor evaluation upon condensate accumulation; (5) DISPENSE — quality threshold met, output pump activated; and (6) RECYCLE — threshold not met, recirculation pump activated.

The quality evaluation logic applies the following acceptance conditions derived from WHO drinking water guidelines: turbidity between 5 and 25 NTU; pH between 6.5 and 9.2; and TDS between 500 and 1500 mg/L. All three conditions must be simultaneously satisfied for water to be classified as PASS and directed to the output port. Any parameter outside its acceptable range triggers a NOT PASS condition and activates the recirculation pump.

The 16×2 LCD display module, interfaced with the Arduino via I²C protocol, provides real-time readout of all four monitored parameters (turbidity, pH, TDS, and temperature) in user-readable format, enabling on-site manual verification of system performance.

4. Mathematical Formulations

The TDS estimation from electrical conductivity follows an empirically validated linear relationship:

$$TDS \text{ (ppm)} = K_c \times EC \text{ (}\mu\text{S/cm)}$$

where K_c is the conversion factor, typically ranging from 0.5 to 0.7 depending on the ionic composition of the solution. For the sensor calibration in this study, $K_c = 0.67$ was applied, consistent with standard TDS meter calibration practices for mixed mineral solutions.

Turbidity measurement using the optical sensor follows the Beer-Lambert principle for light attenuation through a scattering medium:

$$T = I_t / I_0$$

where T is the transmittance ratio, I_t is the transmitted light intensity, and I_0 is the incident light intensity. Turbidity in NTU is inversely proportional to T , with higher suspended

particle concentrations producing greater light scattering and reduced transmittance. The sensor output voltage is mapped to NTU values through a polynomial calibration curve established during initial sensor characterization.

The pH measurement is derived from the Nernst equation governing the electrochemical potential of the pH electrode:

$$E = E_0 - (RT/nF) \times \ln[H^+]$$

where E is the measured electrode potential (mV), E_0 is the standard electrode potential, R is the universal gas constant (8.314 J/mol·K), T is the absolute temperature (K), n is the number of electrons transferred (1 for pH), F is the Faraday constant (96,485 C/mol), and $[H^+]$ is the hydrogen ion activity. The Arduino maps the analog voltage output of the pH sensor module to the pH scale through a linear calibration function established using the two-point buffer calibration procedure.

5. Experimental Setup and Testing Protocol

Water samples were collected from tap water outlets within the College of Engineering and Architecture (CEA) of the Polytechnic University of the Philippines, Sta. Mesa, Manila. The experimental protocol comprised two primary test scenarios: (1) distillation of standard tap water samples; and (2) distillation of intentionally adulterated water samples prepared with acidic solutions and turbidity-inducing agents to test the system's recirculation logic.

For each test run, pre-test measurements of pH, turbidity, TDS, and temperature were recorded prior to introducing water into the system. Post-test measurements were captured by the sensor array in the secondary collection container following distillation. All measurements were evaluated against WHO Guidelines for Drinking-water Quality (4th Edition) threshold values as reference standards. The WHO parameters used are summarized in Table II.

TABLE II. WHO Drinking Water Quality Standards Used as Acceptance Criteria

Parameter	WHO Minimum (Mg/L)	WHO Maximum (Mg/L)	Unit
Turbidity (NTU)	5	25	NTU
pH	7.0 – 8.5	6.5 – 9.2	—
Total Dissolved Solids	500	1500	mg/L
Temperature	—	≤ 25	°C

* Note: The WHO guideline value of 500 mg/L represents the lower palatability threshold; the upper limit of 1500 mg/L is for emergency conditions. Values below 500 mg/L or above 1500 mg/L may indicate unusual treatment outcomes or contamination.

III. RESULTS & DISCUSSION

1. Tap Water Distillation Test

Following a complete distillation cycle applied to the tap water sample sourced from PUP-CEA, the microcontroller's serial monitor reported the following output parameters, as captured from the Arduino IDE console log:

- Turbidity: 4.7 NTU
- pH: 6.8
- TDS: 741 ppm
- Temperature: 26°C

The system issued an 'ALL PASSED – MED Water Level' evaluation flag for the turbidity and pH parameters, confirming that both fell within WHO-prescribed ranges. However, the TDS value of 741 ppm — though within the WHO's absolute maximum of 1500 mg/L — failed to reach the WHO minimum desirable concentration of 500 mg/L, resulting in an overall quality classification that did not meet the full WHO criteria for potable water. Table III presents a comprehensive comparison of pre-distillation and post-distillation parameter values against WHO standards.

TABLE III. Pre-Test and Post-Test Water Quality Parameter Comparison Against WHO Standards

Parameter	Pre-Test (Raw Tap)	Post-Test (Distilled)	WHO Limit	Status
pH	~7.2 (est.)	6.8	6.5 – 9.2	PASS
Turbidity (NTU)	~25 (est.)	4.7	5 – 25	PASS
TDS (ppm)	~750 (est.)	741	500 – 1500	FAIL*
Temperature (°C)	~26	26	≤ 25	NOTE

* FAIL classification for TDS indicates the post-distillation value (741 ppm) exceeds the WHO minimum desirable concentration of 500 mg/L, though it remains within the absolute maximum limit of 1500 mg/L. Additional treatment stages are required to reduce TDS to the recommended minimum.

The turbidity reduction from an estimated baseline of approximately 25 NTU (consistent with the upper acceptable limit for raw tap water) to 4.7 NTU post-distillation represents a significant particulate removal efficiency. This result confirms that the thermal distillation process effectively removed suspended particles and colloidal matter from the water sample, as vapor-phase water transfer inherently excludes non-volatile suspended solids.

The pH value of 6.8 is consistent with expected behavior of thermally distilled water. Pure distilled water theoretically has a pH of 7.0; however, atmospheric CO₂ dissolution during condensation and collection produces carbonic acid, typically depressing the pH to the range of 5.5–6.5 for highly purified distillate. The measured pH of 6.8 indicates residual mineral buffering capacity in the distillate, suggesting incomplete

removal of dissolved mineral species — consistent with the elevated TDS of 741 ppm.

The persistent TDS of 741 ppm following thermal distillation indicates that the single-pass distillation cycle with a nichrome wire heating element was insufficient to achieve complete volatile and dissolved solids removal. This may be attributable to: (1) carry-over of aerosolized droplets during vigorous boiling; (2) insufficient condenser cooling efficiency allowing non-volatile mineral species to co-condense with the water vapor; and (3) the absence of a secondary polishing stage such as reverse osmosis or ion exchange.

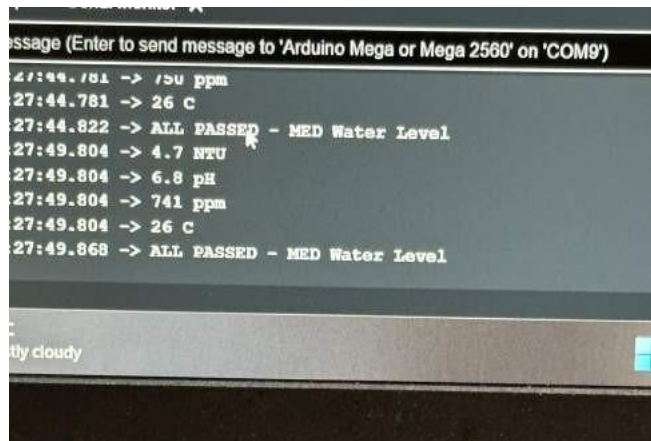


Figure 3: Arduino IDE serial monitor output showing real-time sensor readings during tap water distillation test, confirming turbidity: 4.7 NTU, pH: 6.8, TDS: 741 ppm, Temperature: 26°C, and ALL PASSED status for pH and turbidity

2. Adulterated Water Recirculation Test

To evaluate the closed-loop feedback and recirculation logic, the researchers prepared a water sample consisting of tap water adulterated with an acidic solution and turbidity-inducing agents. Upon evaluation of this sample, the microcontroller correctly identified all monitored parameters as non-compliant and issued a NOT PASSED classification. The recirculation pump was automatically activated, directing the water back to the primary distillation container to undergo an additional treatment cycle.

This result confirmed the functional integrity of the automated decision logic and pump control system. The system accurately distinguished between compliant and non-compliant water samples and executed the appropriate control response without operator intervention. The relay-controlled pump switching demonstrated reliable performance during the testing session, validating the hardware-software integration of the control system.

However, post-recirculation analysis indicated that, while turbidity and pH parameters were brought within acceptable ranges after additional distillation cycles, additional chemical treatment (such as activated carbon filtration or chemical disinfection) would be necessary to achieve TDS reduction to WHO minimum recommended concentrations.

3. System Limitations and Engineering Challenges

Several engineering challenges were identified and documented during the experimental testing phase:

- Extended Distillation Cycle Duration:** The time required to complete a single distillation cycle was identified as a significant operational limitation. The thermal mass of the water volume, combined with the power output of the nichrome wire heating element operating through a TRIAC dimmer, resulted in a distillation rate insufficient for practical continuous water production in household or field deployment scenarios. Precise cycle duration data were not recorded during initial testing; however, qualitative assessment by the research team characterized the distillation time as prohibitively long for rapid water provision. Future iterations should quantify cycle time and benchmark against alternative heating configurations.
- Heating Element Thermal Safety:** The nichrome wire heating element exhibited a propensity for overheating when not fully submerged in water, presenting a risk of element failure or ignition of proximate materials. This was attributed to the absence of a water level interlocking mechanism that would de-energize the heater if the water level in the distillation container fell below the minimum safe submersion threshold. The research team implemented a manual workaround by refilling the primary container during heater-off intervals, but this represents an unacceptable operational limitation for an autonomous system.
- LCD Display Malfunction:** Intermittent display errors were observed with the 16×2 LCD module during testing, resulting in incomplete or garbled parameter readout. This was attributed to potential wiring instability or I²C address conflicts on the shared communication bus. Replacement with a more robust display interface or implementation of I²C signal conditioning is recommended.
- TDS Reduction Insufficiency:** As discussed in Section III-A, single-pass thermal distillation proved insufficient to reduce TDS values to the WHO minimum recommended concentration of 500 mg/L. This is a fundamental limitation of thermal distillation without supplementary demineralization treatment and represents the primary barrier to achieving full WHO potability compliance with the current system architecture.

4. Discussion of System Performance and Significance

The E-Water Distillation System demonstrated successful integration of multi-parameter sensor fusion with automated control logic in a low-cost, portable form factor. The system's ability to correctly evaluate post-distillation water quality in real time and autonomously manage pump routing represents a meaningful advance over conventional manual water quality testing approaches, which require laboratory equipment and trained personnel.

The observed turbidity reduction to 4.7 NTU and pH stabilization at 6.8 within WHO-acceptable ranges validate

the effectiveness of thermal distillation for addressing particulate contamination and pH correction in tap water. These results are consistent with the literature, which documents thermal distillation as a highly effective method for removing suspended solids, biological contaminants, and organic compounds with boiling points above 100°C [1][8].

The failure to achieve TDS compliance in a single distillation pass highlights a well-documented limitation of simple thermal distillation systems: dissolved minerals with boiling points well above water's are effectively excluded from the vapor phase during normal boiling, but trace mineral carry-over through aerosol and droplet entrainment can significantly elevate TDS in the condensate. The incorporation of a multi-stage treatment system — combining pre-filtration, thermal distillation, and post-distillation ion exchange or reverse osmosis polishing — would be expected to reduce TDS to below 500 mg/L and achieve full WHO compliance.

The system's budget-conscious design philosophy, utilizing commercially available components procured through accessible retail channels (Shopee marketplace and Deeco), demonstrates the feasibility of constructing functional water treatment and monitoring systems at minimal cost. This is particularly relevant for deployment in low-resource rural Philippine communities where access to commercial water treatment infrastructure is limited [3][4].

IV. CONCLUSION

A. Conclusion

This paper presented the design, implementation, and experimental evaluation of a Portable E-Water Distillation System controlled by an Arduino Mega 2560 microcontroller. The system integrates pH, turbidity, TDS, temperature, and water level sensors with a nichrome wire heating element, relay-controlled water pumps, and a 16×2 LCD display to automate water distillation and real-time quality assessment.

Experimental testing demonstrated that the system successfully reduced turbidity from estimated tap water baseline levels to 4.7 NTU and maintained a pH of 6.8, both within WHO acceptable ranges for drinking water. The automated quality evaluation and recirculation logic correctly classified non-compliant water samples and activated the appropriate pump routing without operator intervention, confirming the functional integrity of the closed-loop control system.

However, the post-distillation TDS of 741 ppm — exceeding the WHO minimum desirable concentration of 500 mg/L — indicates that single-pass thermal distillation is insufficient to achieve full WHO potability compliance for tap water samples exhibiting moderate dissolved mineral content. Additional treatment stages, including multi-stage filtration and chemical disinfection, are required to produce water that fully meets WHO drinking water quality standards.

The primary contributions of this work are: (1) demonstration of a functional low-cost, portable water distillation system with integrated automated quality monitoring; (2) experimental validation of multi-parameter

sensor integration on an Arduino Mega 2560 platform for water quality assessment; (3) quantitative characterization of the performance and limitations of single-pass nichrome wire thermal distillation relative to WHO standards; and (4) identification of critical engineering challenges — thermal safety, cycle efficiency, TDS reduction, and display reliability — that must be addressed in subsequent system iterations.

B. Recommendations

1. **Solar Power Integration:** The incorporation of photovoltaic power sources would extend system deployability to off-grid rural locations lacking electrical infrastructure, directly addressing the primary deployment context for this system.
2. **Multi-Stage Filtration:** Addition of activated carbon pre-filtration and post-distillation ion exchange or reverse osmosis stages is expected to reduce TDS to below 500 mg/L and address residual chemical contaminants not removed by thermal distillation alone.
3. **Chemical Disinfection:** Integration of ultraviolet (UV) disinfection or chemical dosing (e.g., sodium hypochlorite) would address microbiological contamination risks, providing an additional safety barrier for pathogen removal.
4. **Heater Safety Interlock:** Implementation of a firmware-level water level interlock for the heating element — de-energizing the heater when the primary container level falls below a minimum safe threshold — is essential for autonomous operation without operator supervision.
5. **Custom Enclosure and Heat Dissipation:** Development of a purpose-designed enclosure with integrated heat management (e.g., heat sink, fan-assisted cooling) would improve system reliability and reduce component stress during extended operation cycles.

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REFERENCES

- [1] K. Spandana and V. R. Seshagiri Rao, "Internet of Things (IoT) Based Smart Water Quality Monitoring System," *International Journal of Engineering & Technology*, vol. 7, no. 3.6, p. 259, Jul. 2018, doi: <https://doi.org/10.14419/ijet.v7i3.6.14985>.
- [2] World Health Organization, "Drinking Water," World Health Organization, Sep. 13, 2023. <https://www.who.int/news-room/fact-sheets/detail/drinking-water>
- [3] R. L. Besagas, A. Y. Asoy, M. S. Ceniza, G. D. Leopoldo, N. T. Dael, and R. M. D. Rosario, "Upland and Coastal Freshwater Sources in Misamis Oriental, Philippines," *Mindanao Journal of Science and Technology*, vol. 13, 2015. Available: <https://mjst.ustp.edu.ph/index.php/mjst/article/view/49>
- [4] M. Salehi, "Global Water Shortage and Potable Water Safety; Today's Concern and Tomorrow's Crisis," *Environment International*, vol. 158, no. 0160-4120, p. 106936, Jan. 2022, doi: <https://doi.org/10.1016/j.envint.2021.106936>.
- [5] R. V. Labana, Ma. C. N. M. Borda, R. T. A. Campo, and M. A. V. Ocampo, "The Interplay Between Household Risk Perception of Parasitic Infections and Water, Sanitation, and Hygiene (WASH) Practices: Evidence From an Urban Poor Community in the Philippines," *Cureus*, Apr. 2024, doi: <https://doi.org/10.7759/cureus.57532>.
- [6] K. L. Smalling et al., "Exposures and Potential Health Implications of Contaminant Mixtures in Linked Source Water, Finished Drinking Water, and Tapwater from Public-Supply Drinking Water Systems in Minneapolis/St. Paul Area, USA," Jan. 2023, doi: <https://doi.org/10.1039/d3ew00066d>.
- [7] P. M. Bradley et al., "Tapwater Exposures, Effects Potential, and Residential Risk Management in Northern Plains Nations," *ACS ES&T Water*, vol. 2, no. 10, pp. 1772–1788, Sep. 2022, doi: <https://doi.org/10.1021/acsestwater.2c00293>.
- [8] R. Bogdan, C. Paliuc, M. Crisan-Vida, S. Nimara, and D. Barmayoun, "Low-Cost Internet-of-Things Water-Quality Monitoring System for Rural Areas," *Sensors*, vol. 23, no. 8, p. 3919, Jan. 2023, doi: <https://doi.org/10.3390/s23083919>.
- [9] M. Salehi, "Global Water Shortage and Potable Water Safety; Today's Concern and Tomorrow's Crisis," *Environment International*, vol. 158, no. 0160-4120, p. 106936, Jan. 2022, doi: <https://doi.org/10.1016/j.envint.2021.106936>.